

Topic 2B: Structure
23. know that giant lattices are present in: - i ionic solids (giant ionic lattices) - ii covalently bonded solids, such as diamond, graphite and silicon(IV) oxide (giant covalent lattices) - iii solid metals (giant metallic lattices)
24. know that the structure of covalently bonded substances such as iodine, I ₂ , and ice, H ₂ O, is simple molecular
25. know the different structures formed by carbon atoms, including graphite, diamond and graphene
26. be able to predict the type of structure and bonding present in a substance from numerical data and/or other information
27. be able to predict the physical properties of a substance, including melting and boiling temperature, electrical conductivity and solubility in water, in terms of: - i the types of particle present (atoms, molecules, ions, electrons) - ii the structure of the substance - iii the type of bonding and the presence of intermolecular forces, where relevant